f4_fp: blocked Faugère–Lachartre pair reduction — run report

Run `run-20261016T1720Z` · code commit `1c55f48bb` (branch `claude/f4-f5-algorithm-speed-bvp7qd`) · 2026-10-16

What ran

`examples/gb_cas_export` (release) on the 13 standard and 5 large (`GB_CAS_LARGE=1`) F_p systems, one case per process (`GB_CAS_ONLY`), `RAYON_NUM_THREADS=1`, one F4 run per process. Arms, interleaved per case in each of 3 rounds (`raw/run_ab.sh`):

- **blocked**: default, `F4_FP_FL_BLOCKED=1`;
- **control**: same binary, `F4_FP_FL_BLOCKED=0` (row-by-row `D − C·B'`, the code before this change);
- **msolve 0.6.5** (Ubuntu package), `msolve -g 2 -t 1` on the exported `.ms` file of the same system; process wall time, so it includes msolve's start-up (≈ 5–7 ms, visible on the small cases).

Ours is the in-process F4 time reported by the exporter. Host: Intel Xeon @ 2.10 GHz, 4 vCPU KVM cloud container, shared, unpinned (L0). Runs requested 162, completed 162, no failures.

Correctness

The reduced basis fingerprint (`basis_fp`) is identical between blocked and control in every round for all 18 cases (`raw/ab_full.tsv`). `f4_fp` unit tests 28/28 pass, including the new `blocked_pair_reduction_matches_direct_product`.

Measured values (median of 3 rounds, ms; ratios derived as median / median)

case	blocked	control	msolve	blocked/control	blocked/msolve
`quad_n4_p31`	1.1	0.9	6.8	1.199	0.163
`quad_n5_p31`	1.9	2.1	5.6	0.911	0.334
`quad_n6_p65521`	4.1	5.0	6.7	0.817	0.618
`quad_n7_p65521`	15.1	16.8	17.0	0.898	0.888
`cubic_n3_p29`	1.4	1.3	6.1	1.065	0.231
`cubic_n4_p31`	3.3	4.0	7.2	0.825	0.459
`quart_n3_p31`	2.2	2.4	5.9	0.900	0.372
`overdet_n6_m9_p101`	1.9	1.5	7.0	1.238	0.267
`quad_n8_p65521`	55.0	53.2	47.6	1.033	1.156
`quad_n9_p65521`	255.9	289.8	230.1	0.883	1.112
`cubic_n5_p31`	23.0	25.1	19.1	0.917	1.204
`overdet_n10_m15_p65521`	31.4	27.4	30.0	1.147	1.046
`overdet_n12_m24_p65521`	58.6	57.6	62.8	1.018	0.933
`quad_n10_p65521`	1410.5	1803.0	1798.2	0.782	0.784
`quad_n11_p65521`	9370.3	10548.0	12425.5	0.888	0.754
`cubic_n6_p65521`	261.3	258.4	175.0	1.011	1.493
`quart_n4_p65521`	15.4	14.8	14.7	1.038	1.047
`overdet_n14_m28_p65521`	238.6	262.5	306.0	0.909	0.780

Per-round ranges (`table.tsv`): quad_n10 blocked 1308–1490 vs control 1752–1810; quad_n11 9299–9667 vs 10079–10820; quad_n9 241–294 vs 266–315 (overlap). Cases under ≈ 30 ms vary in both directions by ±20 % between arms and are within this host's noise.

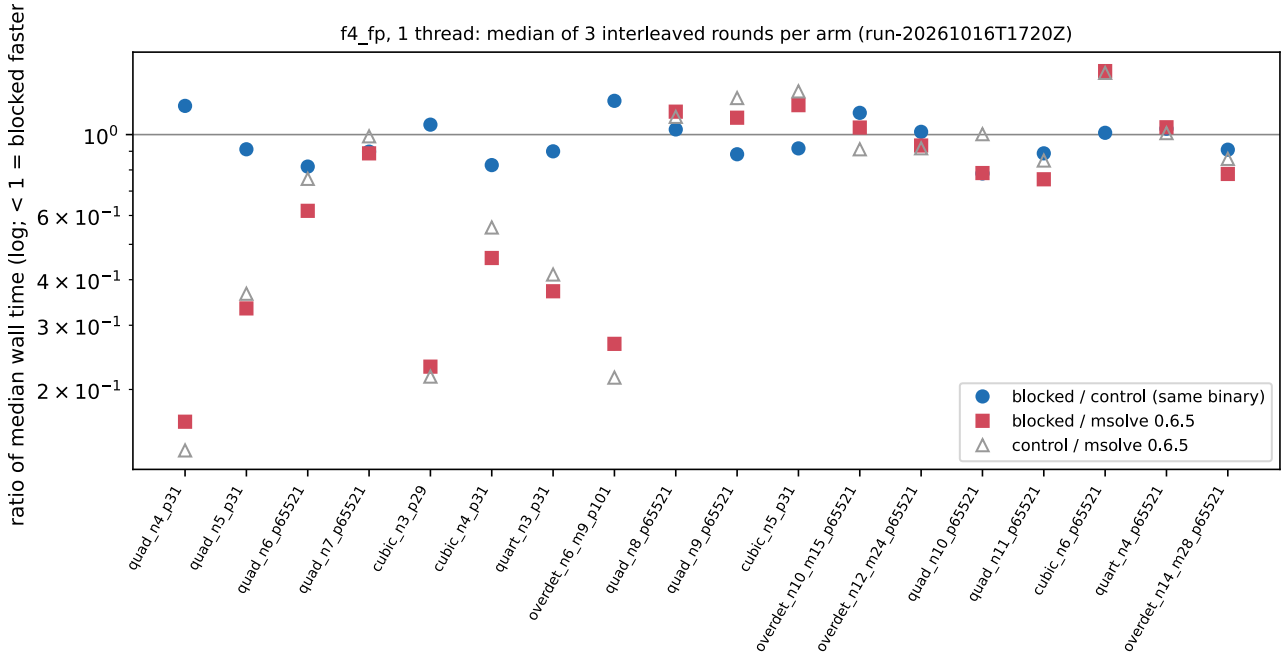


Figure 1: Median ratios per case, 1 thread, n = 3 rounds per arm, run-20261016T1720Z.

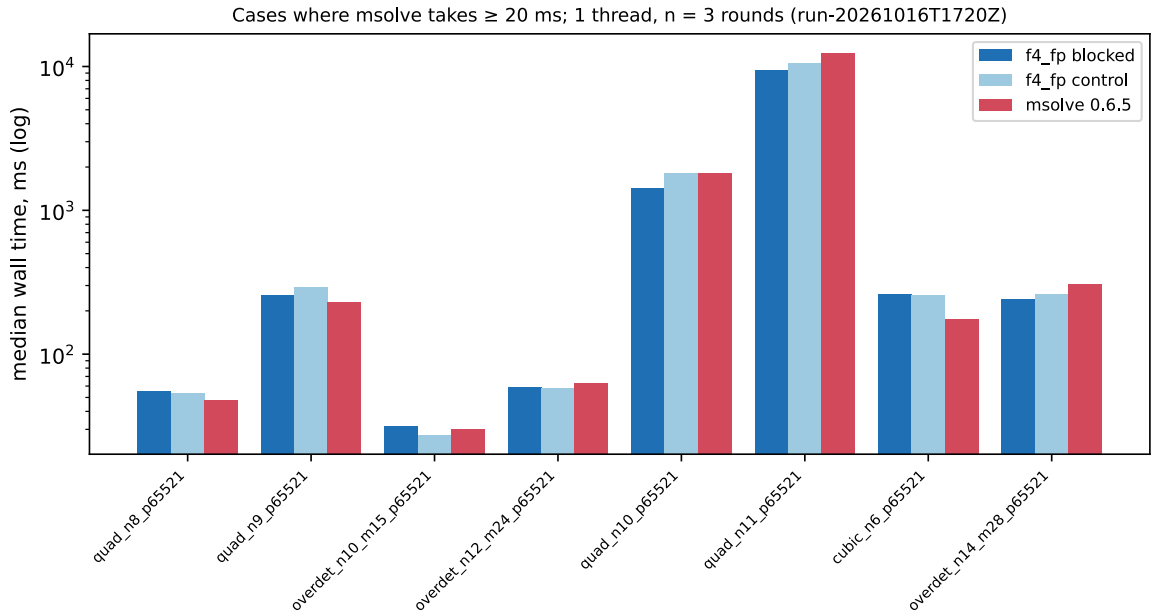


Figure 2: Median wall time for the cases where msolve takes ≥ 20 ms, run-20261016T1720Z.

Observations

- quad_n10 and quad_n11: blocked 0.78× and 0.89× control, per-round ranges disjoint. quad_n9 0.88×, ranges overlap.
- cubic_n6, quart_n4, quad_n8, overdet cases: within noise (0.91–1.15×).
- Against msolve on this host in this run: blocked 0.75–0.78× on quad_n10/n11, 1.11–1.20× on quad_n8/n9/cubic_n5, 1.49× on cubic_n6 (msolve's cubic_n6 ranged 153–204 ms across rounds).
- Not measured here: 4 threads. Exploratory runs before this report (not recorded as a run) showed no difference between arms at 4 threads (quad_n10 ≈ 0.91 s both).
- An extension of the blocking to the `B'` build was also tried exploratorily and showed no gain over this change; it is not in the commit.

Records

`raw/ab_full.tsv` (round, case, arm, ms, basis_fp), `raw/run_ab.sh`, `table.tsv`, `fig_*.svg|png`; plotting source `../../plot.py`.